

Q8		
(a)(i)	α-particles are very strong ionising radiation and hence have very weak penetrating power and would be stopped by a few cm of air. OR Alpha particles could be deflected by air molecules (obscuring the results).	B1
(a)(ii)1	Majority of α-particles passed through with little or no angular deflection. (This suggests that the gold nucleus is made up mostly of empty space, hence the nucleus must be very tiny. Students who stated something to the effect of 'few of the alpha particles underwent deflection or large deflection' would not get the credit. The reason being that the reverse of the statement need not be 'most of the alpha particles were undeflected', it could be that 'most of the alpha particles were absorbed.' This could still imply that the gold nucleus is large in dimensions.)	B1
(a)(ii)2	<u>One of the following:</u> A few α-particles were scattered by large angles. OR A few α-particles backscattered. (The deflections suggest that the gold nucleus is charged (as the alpha-particle is charged, and the back scattering – deflections that are larger than 90° imply that the nucleus is massive)	B1
(b)(i)	$E = \Delta m c^2$ $= [(18.00696 - 18.00568) \times 1.66 \times 10^{-27}] \times (3.0 \times 10^8)^2$ $= 1.91232 \times 10^{-13} \text{ J}$ $= 1.9 \times 10^{-13} \text{ J}$	B1
(b)(ii)	The helium nuclei possessed kinetic energy that can be used for the reaction.	B1
(b)(iii)	The products must also have non-zero kinetic energy after the reaction since the reactants had non-zero total momentum to begin with.	B1

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